

[Process management Based practical]

1. Adam is working in an IT company. He has been given a task to reduce the load of a system by killing some of the processes running in the LINUX operating system. Which commands will he use to complete the given task with the help of the following operation?

● Kill processes by name

```
Chanchal@DESKTOP-GARLJAI MSYS ~
# pacman -S gcc
resolving dependencies...
looking for conflicting packages...

Packages (8) binutils-2.45.1-1 isl-0.27-1 mpc-1.3.1-1 msys2-runtime-devel-3.6.5-1 msys2-w32api-headers-13.0.0.r0.gde62c283f-1
msys2-w32api-runtime-13.0.0.r0.gde62c283f-1 windows-default-manifest-6.4-2 gcc-15.2.0-1

Total Download Size: 72.31 MiB
Total Installed Size: 442.30 MiB

:: Proceed with installation? [Y/n] y
:: Retrieving packages...
msys2-w32api-runtime-13.0.0.r0.gde62c283f-1-x86_64 2.9 MiB 119 KiB/s 00:25 [#####] 100%
binutils-2.45.1-1-x86_64 5.8 MiB 149 KiB/s 00:40 [#####] 100%
mpc-1.3.1-1-x86_64 78.5 KiB 68.0 KiB/s 00:01 [#####] 100%
windows-default-manifest-6.4-2-x86_64 3.0 KiB 971 B/s 00:05 [#####] 100%
msys2-runtime-devel-3.6.5-1-x86_64 5.6 MiB 118 KiB/s 00:48 [#####] 100%
isl-0.27-1-x86_64 718.8 KiB 63.7 KiB/s 00:11 [#####] 100%
msys2-w32api-headers-13.0.0.r0.gde62c283f-1-x86_64 5.7 MiB 88.5 KiB/s 01:06 [#####] 100%
gcc-15.2.0-1-x86_64 51.5 MiB 277 KiB/s 03:10 [#####] 100%
Total (8/8) 72.3 MiB 388 KiB/s 03:11 [#####] 100%
error: failed retrieving file 'isl-0.27-1-x86_64.pkg.tar.zst' from mirror.msys2.org : Operation too slow. Less than 1 bytes/sec transferred the last 30 seconds
(8/8) checking keys in keyring [#####] 100%
(8/8) checking package integrity [#####] 100%
(8/8) loading package files [#####] 100%
(8/8) checking for file conflicts [#####] 100%
(8/8) checking available disk space [#####] 100%
:: Processing package changes...
(1/8) installing binutils [#####] 100%
(2/8) installing isl [#####] 100%
(3/8) installing mpc [#####] 100%
(4/8) installing msys2-runtime-devel [#####] 100%
(5/8) installing msys2-w32api-headers [#####] 100%
(6/8) installing msys2-w32api-runtime [#####] 100%
(7/8) installing windows-default-manifest [#####] 100%
(8/8) installing gcc [#####] 100%
:: Running post-transaction hooks...
(1/1) Updating the info directory file...

Chanchal@DESKTOP-GARLJAI MSYS ~
# gcc --version
gcc (GCC) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
```

● Kill a process based on the process name

```
/C:/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003

Chanchal@DESKTOP-GARLJAI MSYS ~
# cd /C:/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003/

Chanchal@DESKTOP-GARLJAI MSYS /C:/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# sleep 400
Terminated sleep 400

Chanchal@DESKTOP-GARLJAI MSYS /C:/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# ^C

Chanchal@DESKTOP-GARLJAI MSYS /C:/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# |

/C:/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003

Chanchal@DESKTOP-GARLJAI MSYS ~
# cd /C:/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003/

Chanchal@DESKTOP-GARLJAI MSYS /C:/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# ps aux | grep sleep

Chanchal@DESKTOP-GARLJAI MSYS /C:/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# |
```

●Kill a single process at a time with the given process ID

```
chanchal@DESKTOP-GARLJAI MSYS -  
# cd /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003/  
chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# sleep 400  
Terminated sleep 400  
chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# AC  
chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# |  
chanchal@DESKTOP-GARLJAI MSYS -  
# cd /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003/  
chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# ps aux | grep sleep  
chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# |
```

2. Write a program for process creation using C

●Orphan Process

```
GNU nano 8.7  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/types.h>  
  
int main() {  
    int pid = fork();  
  
    if (pid > 0) {  
        sleep(10); // Parent sleeps  
        printf("Parent process running\n");  
    } else {  
        printf("Child process exiting\n");  
    }  
    return 0;  
}
```

```
Retrieving packages...  
msys2-w32api-runtime-13.0.0.r0.gde62c283f-1-x86_64 2.9 MiB 119 KiB/s 00:25 [#####] 100%  
binutils-2.45.1-1-x86_64 5.8 MiB 149 KiB/s 00:40 [#####] 100%  
mpc-1.3.1-1-x86_64 78.5 KiB 68.0 KiB/s 00:01 [#####] 100%  
windows-default-manifest-6.4-2-x86_64 3.0 KiB 571 B/s 00:05 [#####] 100%  
msys2-runtime-devel-3.6.5-1-x86_64 5.6 MiB 118 KiB/s 00:49 [#####] 100%  
isl-0.27-1-x86_64 718.8 KiB 63.7 KiB/s 00:11 [#####] 100%  
msys2-w32api-headers-13.0.0.r0.gde62c283f-1-x86_64 5.7 MiB 88.5 KiB/s 01:06 [#####] 100%  
gcc-15.2.0-1-x86_64 51.5 MiB 277 KiB/s 03:10 [#####] 100%  
Total (8/8) 72.3 MiB 388 KiB/s 03:11 [#####] 100%  
error: failed retrieving file 'isl-0.27-1-x86_64.pkg.tar.zst' from mirror.msys2.org : operation too slow. Less than 1 bytes/sec transferred the last 10 seconds  
(8/8) checking keys in keyring [#####] 100%  
(8/8) checking package integrity [#####] 100%  
(8/8) loading package files [#####] 100%  
(8/8) checking for file conflicts [#####] 100%  
(8/8) checking available disk space [#####] 100%  
:: Processing package changes...  
(1/8) installing binutils [#####] 100%  
(2/8) installing isl [#####] 100%  
(3/8) installing mpc [#####] 100%  
(4/8) installing msys2-runtime-devel [#####] 100%  
(5/8) installing msys2-w32api-headers [#####] 100%  
(6/8) installing msys2-w32api-runtime [#####] 100%  
(7/8) installing windows-default-manifest [#####] 100%  
(8/8) installing gcc [#####] 100%  
:: Running post-transaction hooks...  
(1/1) Updating the info directory file...  
  
chanchal@DESKTOP-GARLJAI MSYS -  
# gcc --version  
gcc (GCC) 15.2.0  
Copyright (C) 2025 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
  
chanchal@DESKTOP-GARLJAI MSYS -  
# nano orphan.c  
chanchal@DESKTOP-GARLJAI MSYS -  
# gcc orphan.c -o orphan  
chanchal@DESKTOP-GARLJAI MSYS -  
# ./orphan  
Child process exiting  
Parent process running  
chanchal@DESKTOP-GARLJAI MSYS -
```

●Zombie Process

M ~

```
GNU nano 8.7
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main() {
    int pid = fork();

    if (pid > 0) {
        sleep(10); // Parent sleeps
        printf("Parent process running\n");
    } else {
        printf("Child process exiting\n");
    }
    return 0;
}
```

M ~

```
Chanchal@DESKTOP-GARLJAI MSYS ~
# nano zombie.c
```

```
Chanchal@DESKTOP-GARLJAI MSYS ~
# gcc zombie.c -o zombie
```

```
Chanchal@DESKTOP-GARLJAI MSYS ~
# ./zombie
Child process exiting
Parent process running
```

```
Chanchal@DESKTOP-GARLJAI MSYS ~
# ps -e1
```

	PID	PPID	PGID	WINPID	TTY	UID	STIME	COMMAND
	1666	1639	1666	15832	pty2	197108	20:39:51	/usr/bin/ps
	1488	1	1488	18032	?	197108	20:17:34	/usr/bin/mintty
I	1657	1489	1657	17304	pty1	197108	20:35:28	/usr/bin/nano
	1638	1	1638	14344	?	197108	20:35:07	/usr/bin/mintty
	1098	1	1098	14008	?	197108	19:41:01	/usr/bin/mintty
	1099	1098	1099	12652	pty0	197108	19:41:02	/usr/bin/bash
	1639	1638	1639	4832	pty2	197108	20:35:07	/usr/bin/bash
	1489	1488	1489	2612	pty1	197108	20:17:34	/usr/bin/bash

```
Chanchal@DESKTOP-GARLJAI MSYS ~
# |
```

3. Create the process using fork () system call.

- Child Process creation
- Parent process creation
- PPID and PID

```
GNU nano 8.7
#include <stdio.h>
#include <unistd.h>

int main() {
    int pid = fork();

    if (pid == 0) {
        // Child process
        printf("Child Process\n");
        printf("PID: %d\n", getpid());
        printf("PPID: %d\n", getppid());
    } else {
        // Parent process
        printf("Parent Process\n");
        printf("PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
    }
    return 0;
}
```

```
Chanchal@DESKTOP-GARLJAI MSYS ~
# nano fork.c

Chanchal@DESKTOP-GARLJAI MSYS ~
# gcc fork.c -o fork

Chanchal@DESKTOP-GARLJAI MSYS ~
# ./fork
Parent Process
Child Process
PID: 1694
PID: 1695
Child PID: 1695
PPID: 1694

Chanchal@DESKTOP-GARLJAI MSYS ~
#
```